

SIP AHEAD: VISUALIZING WATER DEMAND AND FUTURE CONSUMPTION

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INTRODUCTION

Clean water is more than a basic necessity. It's a foundation for good health, education, and economic development. While Malaysia is rich in natural water resources, many regions still experience challenges in accessing consistently treated and safe water. These challenges often stem from a mix of environmental factors, infrastructure gaps, and regional disparities. To ensure clean water security for future generations, it's crucial to understand what contributes to these differences and how we can strengthen our water treatment and distribution systems across the country.

OBJECTIVE



To investigate the clean water access proportion in every state of Malaysia.



To investigate the structure of dams in selected states and their relationship with access to treated water.



To predict future water consumption trends in Malaysia using appropriate predictive model.



To identify possible solutions for long term sustainability of clean water treatment.

METHODOLOGY

Four different methods are done in this research, which are:

- **Data Acquisition**
- **Data Cleaning and Preparation**
- **Data Visualization**
- **Predictive Model for Forecasting and Analysis**

DATA ACQUISITION

Datasets for this research are obtained from various trusted and licensed sources such as open DOSM and myWater Portal. Datasets obtained are combined for further and detailed analysis to seek deeper insights and understanding. Below are the website links for our datasets sources.

- <https://www.dosm.gov.my/portal-main/home>
- <https://data.humdata.org/>
- <https://mywater.gov.my/portal/Modules/SumberAir/Empangan.aspx?Q=4RRbFpCg0pU=>

DATA CLEANING AND PREPARATION

To ensure our research remains centered on Malaysia, filters were applied across all datasets to exclude unrelated data. Null values were removed using Tableau to improve data cleanliness and reliability. For river data, standardization was done by assigning a primary state to each river. This step was taken to enhance the accuracy and consistency of geographic information, especially where rivers cross state boundaries.

DATA VISUALIZATION

All datasets were imported into Tableau for visualization purposes. Various visual formats including graphs, charts, and maps were generated to help interpret trends and explain key insights. Interactive dashboards were created to provide a comprehensive overview of the current situation and uncover relationships between variables. Finally, a data story was developed by sequencing these dashboards to present a clear and cohesive narrative of the findings.

PREDICTIVE MODEL FOR FORECASTING AND ANALYSIS

To forecast clean water consumption in Malaysia, we used two different models: Double Exponential Smoothing and Linear Regression. Double Exponential Smoothing, also known as Holt's Linear Exponential Smoothing, is chosen because our dataset clearly shows a trend over time but doesn't show any seasonal patterns. Linear regression on the other hand provides prediction based on previous trends and used to investigate the relationship between dependent and independent variables. The predictions for both models are implemented using Python. We used the Streamlit library to build a simple, interactive application that runs the models and displays the forecast in a user-friendly way.

The Flow - The Cycle



Objective 1: To investigate the clean water access proportion in every state of Malaysia.

WATER RESOURCES IN MALAYSIA



TOTAL RIVERS

189

TOTAL DAMS

27

Water Resource Breakdown:

Rivers (80.8%)

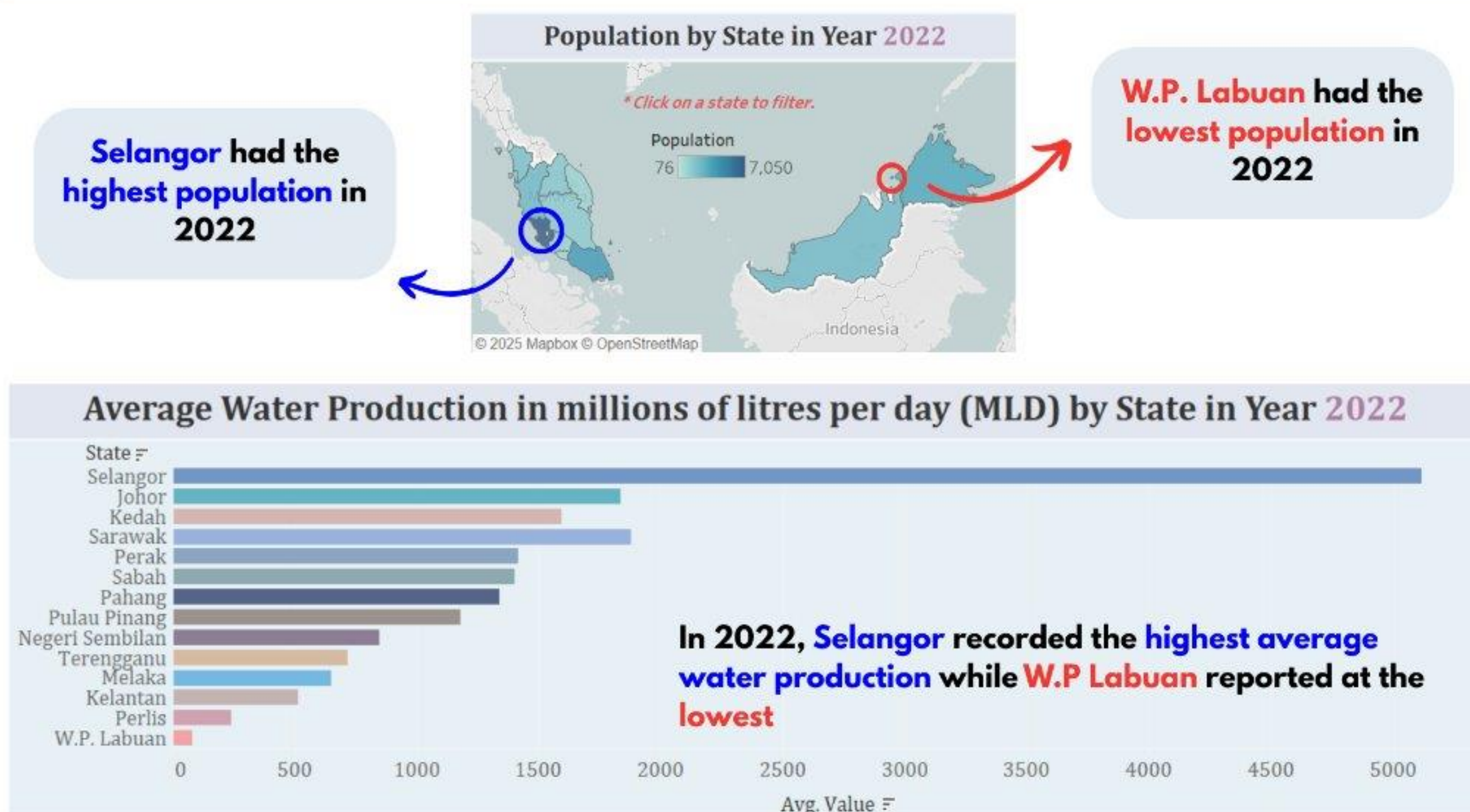
Dams (17.9%)

Groundwater (1.2%)

Selangor has the most abundant water resources while Perlis has the least

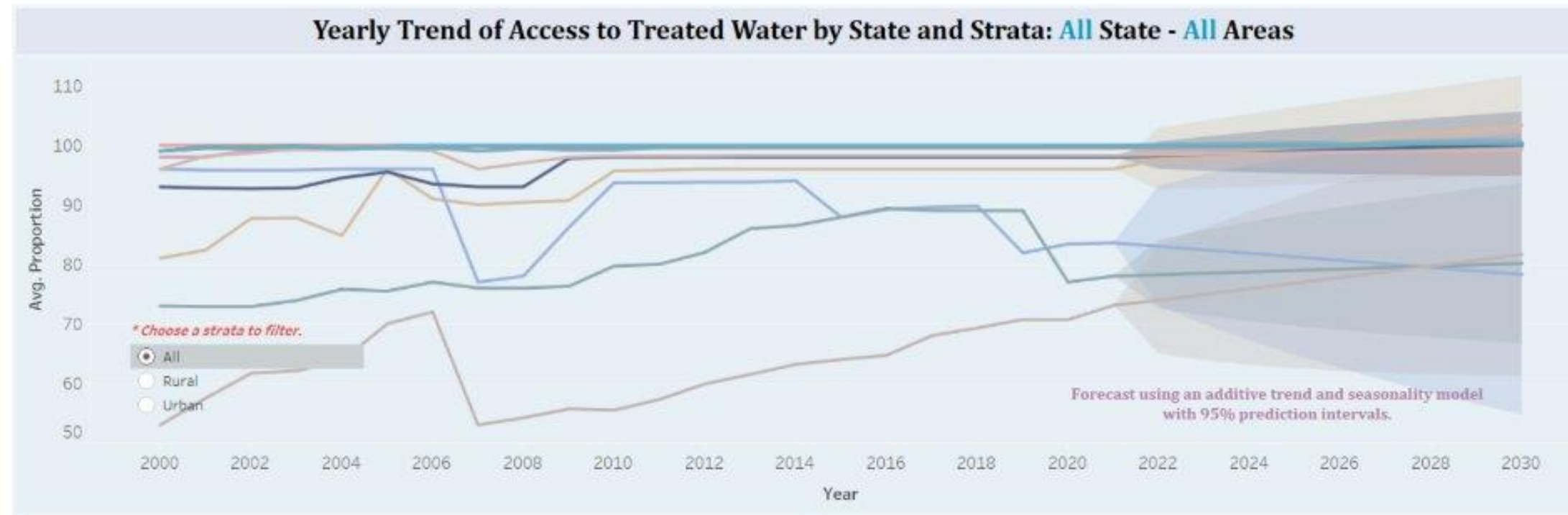
Objective 1: To investigate the clean water access proportion in every state of Malaysia.

CLEAN WATER DEMAND & ACCESS



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CLEAN WATER DEMAND & ACCESS



Based on the graph, Kelantan has significantly low access of treated water yearly followed by Sabah and Sarawak which the access is less than 90%. Clean water access in these 3 states are expected to be increased in the future according to the forecast.

<90% Access

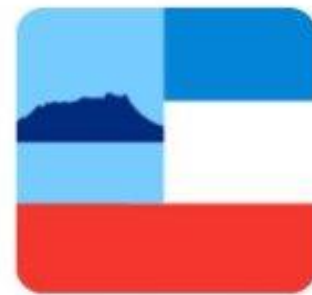


>90% Access



Objective 2: To investigate the structure of dams in selected states and their relationship with access to treated water.

COMPARING THE CHARACTERISTICS OF DAMS IN SARAWAK, SABAH AND KELANTAN WITH SELANGOR



VS



WHY SELANGOR?

- Highest population in 2022
- High access to treated water (>90%)
- Highly developed state

Objective 2: To investigate the structure of dams in selected states and their relationship with access to treated water.

COMPARISON OF DAMS CHARACTERISTICS

Tall Dam Height + Large Reservoir Area + Large Reservoir Capacity = High Access To Treated Water?

State	SARAWAK	SABAH	KELANTAN		SELANGOR	
Dams	Batang Ai	Tenom Pangi	Pergau	Bukit Kwong	Batu	Klang Gates
Dam Height (m)	110.0	83.0	75.0	7.6	44.0	37.0
Reservoir Area (km ²)	24.0	152.0	4.3	4.0	2.5	74.5
Reservoir Capacity (million m ³)	750.0	12.9	62.6	36.6	25.1	14.3

Yellow = Highest Value Red = Lowest Value

Despite having large or tall dams, Sarawak, Sabah and Kelantan still have low access to treated water, suggesting that infrastructure alone does not guarantee high clean water access.

Objective 2: To investigate the structure of dams in selected states and their relationship with access to treated water.

COMPARISON OF DAMS CHARACTERISTICS

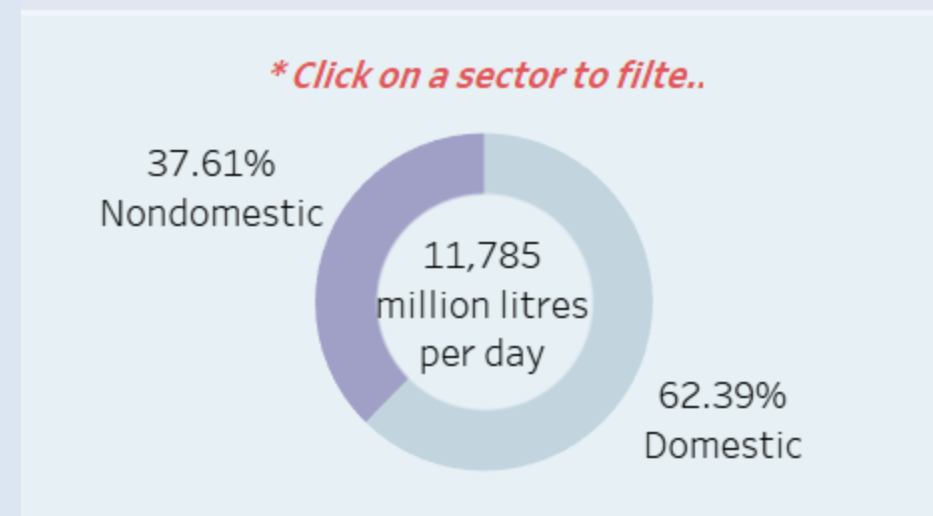


Besides the physical characteristics of dams, function of dams is another aspect to be discussed. Among the six dams, only three dams are built for water supply which is Bukit Kwong, Batu and Klang Gates. In terms of flood control only Batu dam in Selangor helps manage flood which reflects the severe flood condition in the other three states. While for irrigation, only Bukit Kwong has irrigation to supply water for farming.

Objective 3: To predict future water consumption trends in Malaysia using appropriate predictive model.

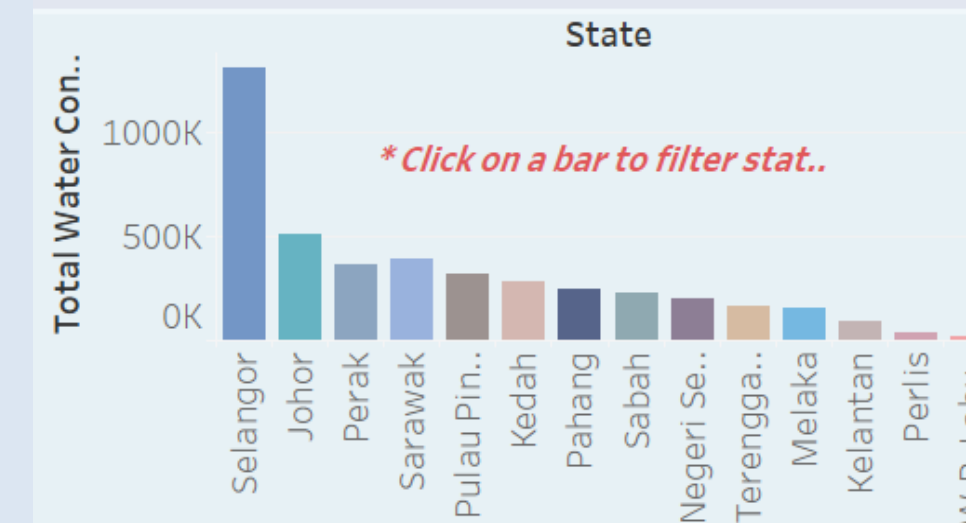
WATER CONSUMPTION IN MALAYSIA

Daily Water Consumption by Sector – Year 2022



Most water consumption in 2022 was for **domestic** purposes, indicating a higher demand from households compared to industries or agriculture. Malaysians heavily depended on clean water for daily personal and residential uses like drinking, cooking and cleaning.

Annual Water Consumption by State – Year 2022



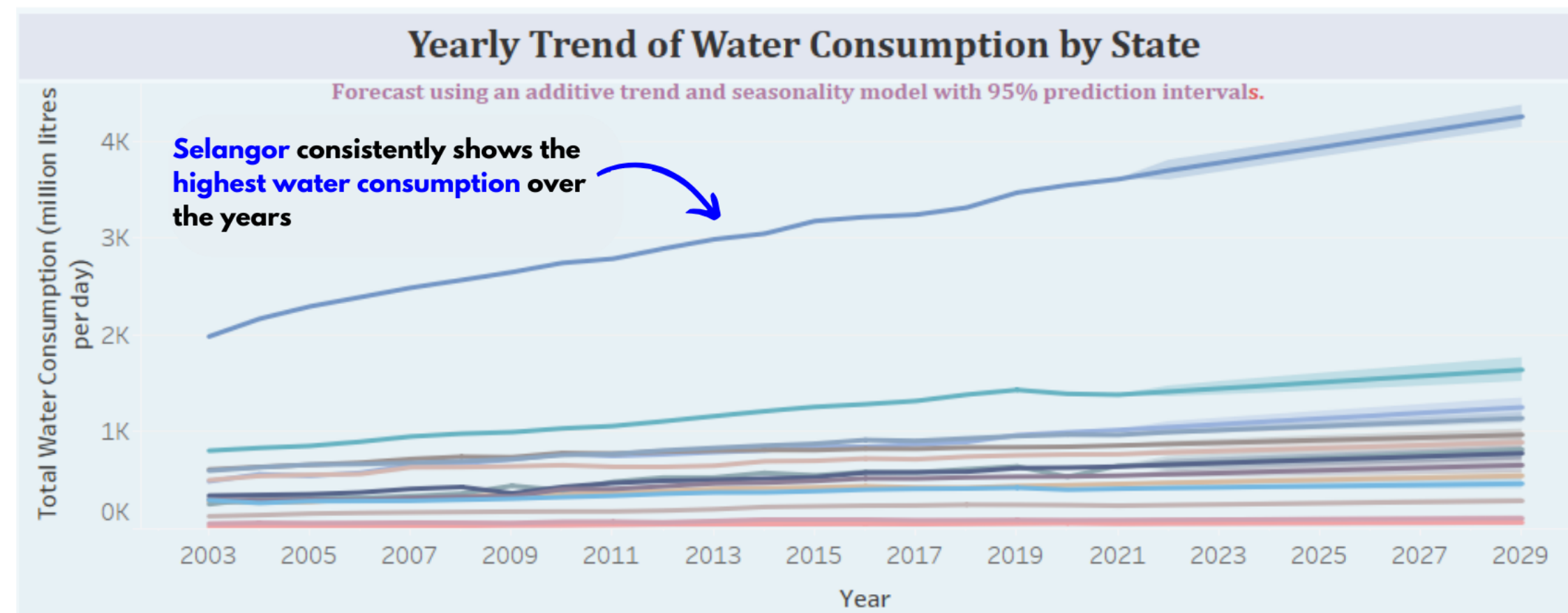
Selangor had the **highest water consumption** in 2022, far exceeding the other states. This is expected as Selangor is the most populated and industrialized state.

W.P. Labuan with the smallest population had the **lowest water consumption**.

This suggests that states with larger population and more development consume more water.

Objective 3: To predict future water consumption trends in Malaysia using appropriate predictive model.

WATER CONSUMPTION IN MALAYSIA



Objective 3: To predict future water consumption trends in Malaysia using appropriate predictive model.

WATER CONSUMPTION IN MALAYSIA

Sum of Value

Color State	Initial 2022	Change From Initial 2022 – 2029	Seasonal Effect		Contribution Trend Season		Quality
W.P. Labuan	54 ± 5	11	2029 0	2029 0	100.0%	0.0%	Ok
Terengganu	474 ± 21	74	2029 0	2029 0	100.0%	0.0%	Ok
Selangor	3,700 ± 101	556	2029 90	2029 90	100.0%	0.0%	Good
Sarawak	1,049 ± 54	206	2029 -10	2029 -10	100.0%	0.0%	Ok
Sabah	665 ± 79	154	2029 -11	2029 -11	100.0%	0.0%	Ok
Pulau Pinang	874 ± 28	96	2029 0	2029 0	100.0%	0.0%	Ok
Perlis	93 ± 12	14	2029 0	2029 0	100.0%	0.0%	Poor
Perak	998 ± 30	144	2029 -1	2029 -1	100.0%	0.0%	Ok
Pahang	663 ± 47	117	2029 -4	2029 -4	100.0%	0.0%	Ok
Negeri Sembilan	563 ± 37	95	2029 0	2029 0	100.0%	0.0%	Poor
Melaka	421 ± 25	46	2029 0	2029 0	100.0%	0.0%	Ok
Kelantan	245 ± 14	43	2029 0	2029 0	100.0%	0.0%	Ok
Kedah	789 ± 39	104	2029 -1	2029 -1	100.0%	0.0%	Poor
Johor	1,416 ± 55	225	2029 0	2029 0	100.0%	0.0%	Ok

All forecasts were computed using exponential smoothing.

Sum of Value

Color State	Level	Model Trend Season	Quality Metrics					Smoothing Coefficients		
			RMSE	MAE	MASE	MAPE	AIC	Alpha	Beta	Gamma
W.P. Labuan	Additive	Additive Additive	2	2	0.74	4.5%	45	0.467	0.000	0.470
Terengganu	Additive	Additive Additive	11	8	0.63	2.3%	103	0.500	0.000	0.500
Selangor	Additive	Additive Additive	52	30	0.34	1.2%	162	0.065	0.469	0.000
Sarawak	Additive	Additive Additive	28	23	0.68	3.3%	138	0.205	0.000	0.411
Sabah	Additive	Additive Additive	41	29	0.69	6.6%	153	0.134	0.000	0.245
Pulau Pinang	Additive	Additive Additive	15	12	0.79	1.6%	114	0.416	0.000	0.419
Perlis	Additive	Additive Additive	6	5	1.01	6.1%	81	0.450	0.000	0.454
Perak	Additive	Additive Additive	15	11	0.52	1.5%	116	0.235	0.000	0.441
Pahang	Additive	Additive Additive	24	17	0.73	4.1%	133	0.131	0.000	0.342
Negeri Sembilan	Additive	Additive Additive	19	13	0.88	3.5%	123	0.478	0.000	0.481
Melaka	Additive	Additive Additive	13	9	0.76	2.8%	108	0.500	0.000	0.500
Kelantan	Additive	Additive Additive	7	6	0.79	3.0%	87	0.500	0.000	0.500
Kedah	Additive	Additive Additive	20	15	0.83	2.3%	126	0.237	0.000	0.443
Johor	Additive	Additive Additive	28	20	0.54	1.8%	139	0.500	0.000	0.500

This forecast uses exponential smoothing (ETS) to predict future values by assuming a consistent linear growth over time and fixed seasonal patterns. The yearly trend of water consumption is forecasted using an additive trend and seasonality model with 95% prediction interval.

Selangor continues to show rapid growth in water consumption over the years due to its increasing population and development. Other states show slow to moderate growth with minor fluctuations, likely due to the states having smaller populations and lesser urbanization.

This forecast helps to predict future water consumption trends across the states in Malaysia, enabling efficient planning and resource management.

Objective 3: To predict future water consumption trends in Malaysia using appropriate predictive model.

CLEAN WATER CONSUMPTION PREDICTIVE MODEL



**Scan the QR code to explore a
predictive model of water
consumption trends in Malaysia**

*See the trends,
understand the needs,
and be part of the
solution!*



Objective 3: To predict future water consumption trends in Malaysia using appropriate predictive model.

CLEAN WATER CONSUMPTION PREDICTIVE MODEL



The predictive model applied machine learning knowledge. The water consumption is predicted using double exponential smoothing and linear regression. Double exponential smoothing is chosen as a prediction method because it does not involve seasonality of data which best suit to our data while linear regression predict water consumption based on the trends. Based on the graph, we can see that total water consumption in Malaysia will increase yearly. This predictive model raise the awareness of government to improve the infrastructure of clean water supply to fulfil the demand of people. This predictive model aligns to the third objective which is identifying possible solution for long term clean water access.

Model Equation:

Level update: $l_t = \alpha y_t + (1 - \alpha)(l_{t-1} + b_{t-1})$

Trend update: $b_t = \beta(L_t - L_{t-1}) + (1 - \beta)b_{t-1}$

Holt's Forecast: $\hat{y}_{t+h} = l_t + hb_t$

for h periods into the future

Objective 4: To identify possible solutions for long term sustainability of clean water treatment.

RECOMMENDED WATER TREATMENT SOLUTION

WATER TREATMENT RECOMMENDATIONS

Method	Electrocoagulation	Photocatalysis
Application	Industrial use	Household use
Treatment Type	Wastewater treatment	
How it works	Uses electric current to dissolve or remove pollutants from water	Uses sunlight and photocatalysts to remove pollutants from water
Cost	Low to moderate initial and operating cost, as it includes simple equipment and electrode maintenance	Moderate to high initial cost, with low to moderate operating cost if solar energy is used. Overall cost may vary depending on system design

SUMMARY

This research has provided an overview of clean water supply in Malaysia. The objectives of the research is achieved. Based on the analysis, clean water access proportion is above 90% for all states in Malaysia except Sarawak, Sabah, and Kelantan. Although these three states have tall and large dams, it does not guarantee clean water supply to their people as the dams built might have other functions such as supplying water specifically for farming. Besides that, the dams built in Sarawak, Sabah, and Kelantan does not help in managing flood control which contribute to the lack of access of clean water. A predicting model has been built to predict water consumption of Malaysians by state which can be a reference for the policymakers. Also, several water treatment solutions such as electrocoagulation and photocatalysis are suggested to achieve sustainable cities development by increasing the proportion of people that have access to clean water.

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